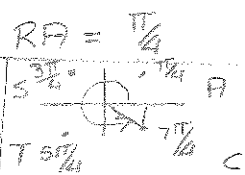
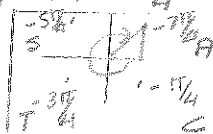


D - Trigonometric Functions

1(a) $\sin(\frac{7\pi}{4}) = -\frac{1}{\sqrt{2}}$

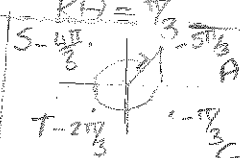


(b) $\operatorname{cosec}(-\frac{7\pi}{4}) = \operatorname{cosec}(\frac{\pi}{4})$ RA is $\frac{\pi}{4}$
 $= \frac{1}{\sin(\frac{\pi}{4})} = \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2}$

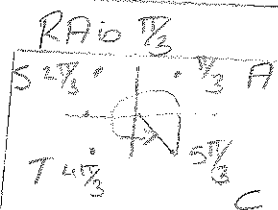


(c) $\sin(35\pi) = \sin(2\pi \times 17 + \pi) = \sin \pi = 0$

(d) $\cos(-\frac{5\pi}{3}) = \cos(\frac{\pi}{3}) = \frac{1}{2}$ RA is $\frac{\pi}{3}$

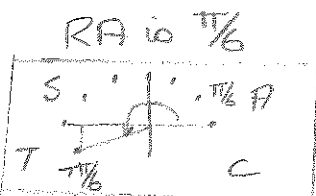


(e) $\sec(\frac{5\pi}{3}) = \sec(-\frac{\pi}{3}) = \frac{1}{\cos(-\frac{\pi}{3})} = \frac{1}{\cos \frac{\pi}{3}} = \frac{1}{\frac{1}{2}} = 2$

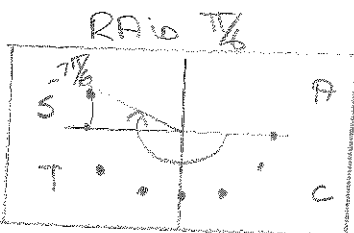


(f) $\sec(\frac{15\pi}{2}) = \sec(\frac{\pi}{2}) = \frac{1}{\cos \frac{\pi}{2}}$ not defined!
 $\cos \frac{\pi}{2} = 0$

(g) $\tan(\frac{7\pi}{6}) = -\frac{1}{\sqrt{3}}$



(h) $\cot(-\frac{7\pi}{6}) = -\sqrt{3}$



(i) $\cos(-\frac{15\pi}{3}) = \cos(-5\pi) = \cos(-\pi) = -1$

2) a) $\sin \theta = \frac{7}{25}, \theta \in (\frac{\pi}{2}, \pi)$



$x^2 + 7^2 = 25^2$

$x^2 = 625 - 49$

$= 576$

$x = \sqrt{576} = \pm 24$

$\therefore x = -24$

$\cos \theta = \frac{24}{25}$

$\tan \theta = \frac{7}{-24} = -\frac{7}{24}$

$\operatorname{cosec} \theta = \frac{1}{\sin \theta} = \frac{25}{7}$

$\cot \theta = -\frac{24}{7}$

(b) $\cos \theta = -\frac{4}{5}, \theta \in (\frac{\pi}{2}, \pi)$



$y = 25 - 16 = 9$

$y = 3$

$\sin \theta = \frac{3}{5}$

$\tan \theta = -\frac{3}{4}$

$\sec \theta = -\frac{5}{4}$

$\operatorname{cosec} \theta = \frac{5}{3}$

(c) $\tan \theta = \frac{5}{12}, \theta \in (\pi, \frac{3\pi}{2})$



$\sin \theta = -\frac{5}{13}$

$\cos \theta = -\frac{12}{13}$

$\cot \theta = \frac{12}{5}$

$\sec \theta = -\frac{13}{12}$

$r = \sqrt{(12)^2 + (5)^2}$

$= \sqrt{169}$

$= 13$

(d) $\cos \theta = -1, \theta \in (\frac{\pi}{2}, \frac{3\pi}{2})$

$\therefore \theta = \pi$

$\sin \theta = 0$

$\tan \theta = \frac{\sin \theta}{\cos \theta} = 0$

$\cot \theta$ not defined

$\operatorname{cosec} \theta$ not defined